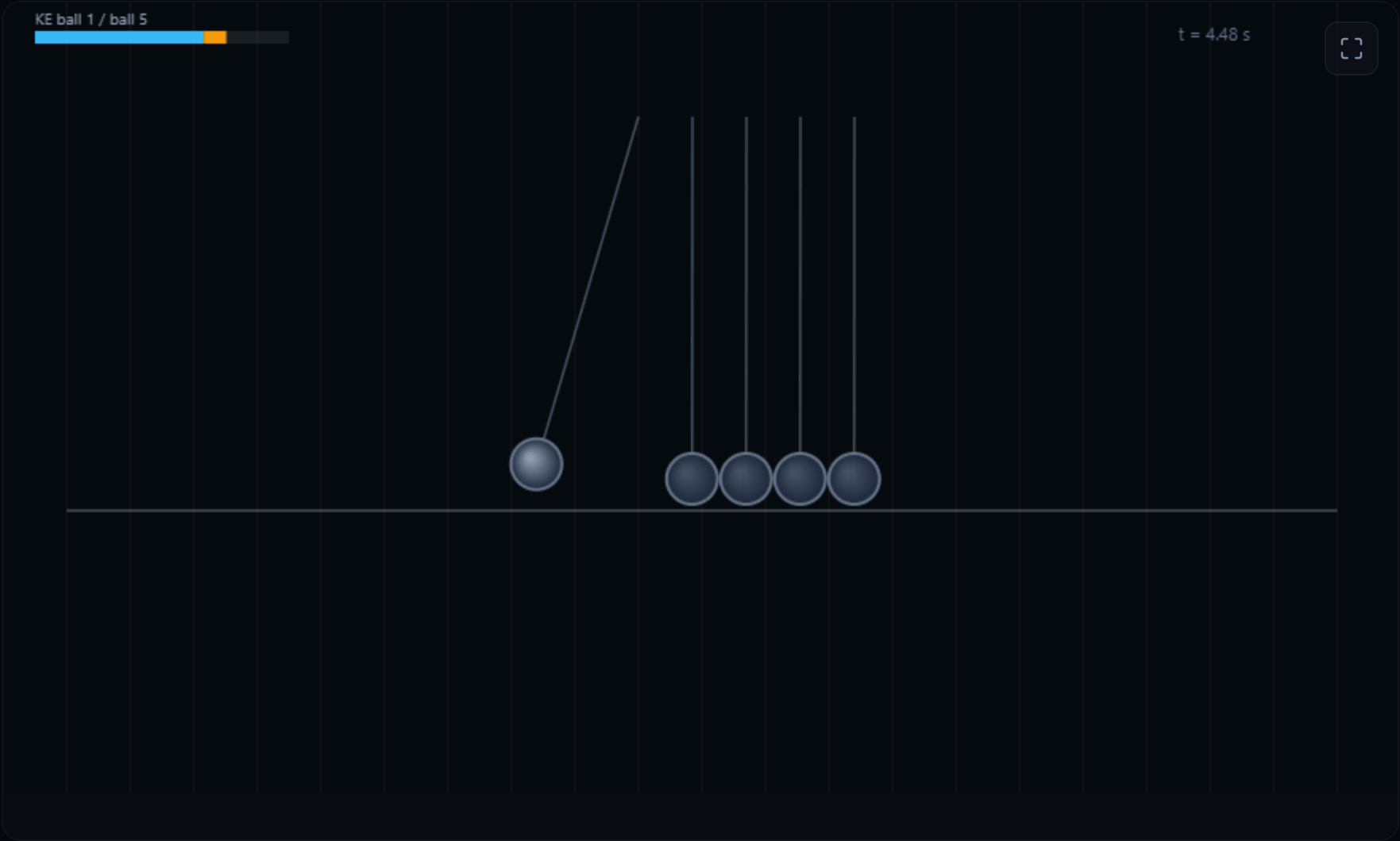
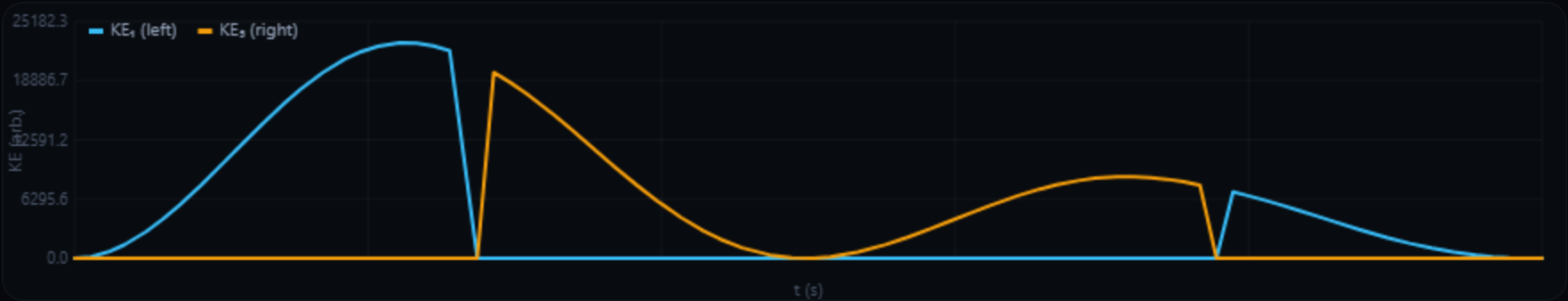


Newton's Cradle

Conservation of momentum and energy visualized with swinging balls.



LIVE GRAPHS



LAUNCH

Initial speed of left ball

340 (arb.)

Air / hinge damping

0.2

Five equal spheres in a line. Tap Release to start — until then the cradle is at rest. The left ball strikes the pack; elastic collisions swap speeds along the chain so the last ball pops out, then the pattern reverses.

DISPLAY

☒ Energy strip (KE₁ vs KE₅)

Release

Reset

SHORTCUTS

- Space or Enter — release swing
- R — reset

MEASURED VALUES

t	4.43 s
KE (ball 1)	11
KE (ball 5)	10

About this model

Newton's cradle is the desk toy of suspended metal spheres that collide in a line. Ideal elastic collisions between identical spheres transfer momentum and energy in a way that launches one outgoing sphere when one incoming sphere strikes a stationary chain—an illustration of one-dimensional impulse propagation and approximate Hertzian contact mechanics. Real cradles deviate: dispersion in the balls and supports, slight misalignment, internal modes, and inelasticity cause secondary motion and imperfect single-sphere ejection. The simulator typically treats spheres as point masses or rigid bodies with a chosen coefficient of restitution, sometimes including small angles for visual flair but keeping the pedagogy one-dimensional. Students compare momentum-conserving limits with energy loss, discuss why the *last* ball pops out in the ideal story, and connect the demo to collision lessons in 1D and 2D simulators elsewhere in the catalog.

Who it's for: High school mechanics classes reinforcing impulse, momentum conservation, and coefficients of restitution after introducing Newton's laws.

KEY TERMS

- Impulse
- Momentum conservation
- Elastic collision
- Coefficient of restitution
- Hertzian contact
- 1D chain
- Dispersion
- Newton's cradle

How it works

Newton's cradle idealizes equal masses in a line with elastic impacts. In 1D, two equal spheres in a head-on elastic collision exchange velocities. Here a small chain propagates that impulse; light damping models air and hinge losses so motion eventually stops.

Key equations

Elastic, equal masses: $v_1' = v_2$, $v_2' = v_1$ (head-on)

$p = \sum m_i v_i$ conserved in elastic collisions

Frequently asked questions

If momentum is conserved, must kinetic energy be conserved too?

No. Momentum can be conserved with partial energy loss to heat and sound. The textbook equal-mass elastic swap is a special case with $e = 1$.

Why do real cradles sometimes launch two balls?

Small gaps, angular wobble, and imperfect timing break the ideal simultaneous-contact assumption, redistributing momentum across neighbors.

Is this the same math as the 1D collisions lab?

Yes in spirit: sequential pairwise collisions with restitution laws. The cradle emphasizes wave-like impulse transfer through multiple contacts.

Does angular momentum matter here?

If spheres twist or strings apply torques, yes. Many didactic models freeze rotation and treat line-of-centers impacts only.

Related simulators

Continue with similar topics in this category — or [all 85 in Classical Mechanics](#).

[View category](#) →

SCHOOL

1D Collisions

▶ Launch Simulator

Elastic and inelastic collisions with momentum and energy tracking.

SCHOOL

2D Collisions

▶ Launch Simulator

Billiard-ball style collisions with adjustable angles.

NEW

SCHOOL

Wall Bounce

▶ Launch Simulator

2D box: frictionless walls, restitution e , optional gravity. Trails and kinetic energy graph.

NEW

SCHOOL

Oblique Wall Impact

▶ Launch Simulator

One tilted segment: e , angles to normal n , open boundaries vs closed Wall Bounce.

NEW

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Pendulum Collision

▶ Launch Simulator

Two bobs: hit along normal, e elastic; θ'' between hits vs 1D collisions.

FEATURED

SCHOOL

Energy Conservation

▶ Launch Simulator

Ball on a roller coaster track with real-time KE, PE, and total energy bars.